D DigestAnything

RALPHTHON SEOUL #2

SOLO BUILD

DigestAnything

Any source → **See it. Test it. Prove it.**

Paste a URL, YouTube link, or PDF → get 5 AI-powered learning experiences

Builder
Jaeho Shin

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GitHub
jays0606/DigestAnything

PROBLEM

Passive Learning Doesn't Stick

80%

of content forgotten within 24 hours

— Ebbinghaus Forgetting Curve

Every day, millions read articles, watch videos, skim papers. **Feels productive. Almost nothing retained.**

The Real Pain

- ✕ No active recall — just highlighting and scrolling
- ✕ No self-testing — can't explain what you just read
- ✕ No reinforcement — content consumed once, forgotten forever

> *"I watched a 1-hour Karpathy lecture. Next day, I couldn't explain a single concept to a colleague."*

SOLUTION

Paste Anything → Learn Everything

URL, YouTube, PDF → 5 experiences grounded in learning science. Not AI gimmicks.

Paste URL → AI analyzes → **Overview loads instantly** → Quiz, Cards, Podcast generate in parallel → **Tutor says "explain it"**

🧠 Overview

Summary + interactive mindmap. See the whole structure at once.

Dual Coding — visual + verbal = 2x retention

❓ Quiz

10 MCQs. Wrong answer → explanation. Brain must output to learn.

Active Recall — output > input

🃏 Cards

12-15 flashcards. Review right before you forget.

Spaced Repetition — beat the forgetting curve

🎙️ Podcast

2-host AI podcast. Learn on your commute.

Elaborative Interrogation — why? how?

🎓 Tutor (Killer)

"Explain this concept." Finds gaps. Never gives the answer.

Feynman + Desirable Difficulty — brain must sweat

💬 Chat

Ask anything about the source. Always available.

On-demand reinforcement

ARCHITECTURE

One Ingest, Parallel Everything

Pipeline

1 **Source Input**
URL / YouTube / PDF

↓

2 **Gemini 3 Flash Analysis**
→ context.json (concepts, hierarchy, insights)

↓

asyncio.gather() — ALL PARALLEL

Visual | Quiz | Cards | Podcast

Stack

Frontend	Next.js 14 + Tailwind
Backend	FastAPI (async)
AI	Gemini 3 Flash + TTS
Design	Google Stitch
Diagrams	Markmap
Agent	Claude Code + oh-my-claudecode

RALPH SETUP

I Wrote the Spec. The Agent Built Everything.

The Ralph Rule: Don't Touch the Laptop

1. **Write Spec + Tests First**
   895-line SPEC.md, 504-line verified API ref, 17 test files

2. **Claude Code /autopilot Builds**
   4 rounds: BE + FE in parallel. Commits after each round.

3. **Agent Test Loop (Opus 4.6)**
   Test → Fail → Agent reads error + code → Fixes → Retest

4. **Playwright E2E + Audio QA**
   Browser tests every tab + Gemini validates podcast audio

Generator-Evaluator (GAN-Inspired)

From Anthropic's **Harness Design** blog (this week): separate the builder from the critic. Self-evaluation bias means you can't trust code to judge itself.

Generator: oh-my-claudecode writes all code

Evaluator: 17 test scripts verify independently

Self-heal: Claude Agent SDK (Opus) reads failures, patches code, retests

What I Actually Did

Wrote SPEC.md → walked away → agent built, tested, fixed, committed. XX commits. XX,XXX lines. Zero manual code.

DEMO

Let's Digest Something

Try it with

📝 Anthropic Engineering Blog

anthropic.com/engineering/harness-design

🎬 Karpathy YouTube Talk

youtube.com/watch?v=kwSVtQ7dziU

~10s	5	100%
Overview loads	Learning tabs	Science-backed

IMPACT

The Last Mile of Education

Access is solved.

MIT OpenCourseWare. YouTube. arXiv. The best lectures in history are **free**.

Understanding is not.

A professor can't ask every student *"explain it back to me."* There's 300 of them in the lecture hall.

AI can.
Every student. Every concept. Infinite patience.

Who Benefits

Students
Paste lecture PDF → quiz yourself → tutor catches gaps → actually pass the exam

Developers
Paste technical blog → mindmap architecture → quiz edge cases → explain to teammates

Lifelong Learners
Paste YouTube talk → podcast for commute → flashcards for spaced review

Educators
Generate quiz + cards from any material instantly. AI tutor handles 1-on-1.

PROGRESS

Current Status

AGENT RUNTIME

5h+

Autonomous build time

COMMITS

20+

All agent-authored

LINES OF CODE

5K+

Python + TypeScript

Overview	Quiz	Cards	Podcast	Tutor
Done	Done	Done	Done	Polishing

Human contribution: SPEC.md (895 lines) + test definitions. Everything else: autonomous agent.